

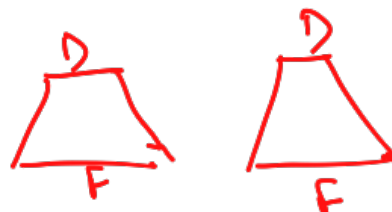
Question 1: How many parameters does a model with $D = 4096$, $F = 4 \cdot D$, $V = 32,000$, and $L = 64$ have? What fraction of these are attention parameters? How large are our KV caches per token? You can assume $N \cdot H = D$ and multi-head attention with int8 KV's.

layers →

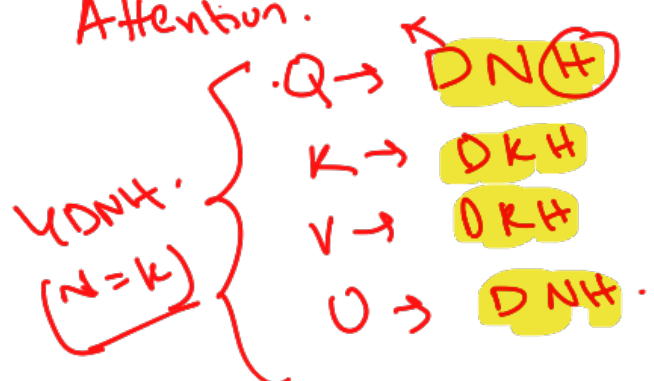
parameters

MLP

3DF.



Attention.



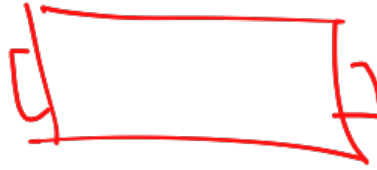
$N=K$

Multi head attention



1 expert 2D

Embedding = DV + DV → language modeling head.
tokens → d-model



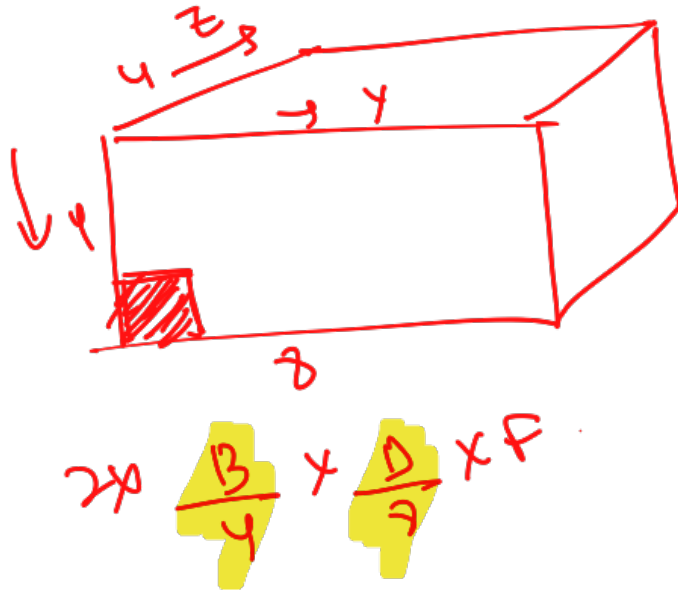
$$\frac{\text{Attention}}{\text{total \# parameters}} = \frac{4DNH}{3DF + 4DNH + 2DV + 2D}$$

② KV cache

$$2 \times L \times N \times H$$

K&V

Question 2: How many total FLOPs are required to perform $A[B_X, D_Y] *_{\underline{D}} W[\underline{D}_Y, F]$ on $\{X: 4, Y: 8, Z: 4\}$? How many FLOPs are performed by each TPU?



Question 3: How many FLOPs are involved in performing

$$A[I, J, K, L] * B[I, J, M, N, O] \rightarrow C[K, L, M, N, O]?$$

2 k L M N D I J O

Question 4: What is the arithmetic intensity of grouped multi-query attention (ignoring the

(*) Q/K/V/O projections)? Give the answer as a function of the Q and KV sequence lengths T and S and the multi-query factor G . At what context length is attention FLOPs-bound? Given the HBM bandwidth of our TPUs, plot the effective relative cost of attention to the FFW block as the context length grows. Hint: assume we're using an efficient attention implementation that doesn't do any unnecessary reads/writes. Consider both the limits where $T = S$ and $T \ll S$.

$$X = BTD \rightarrow N$$

$$W_Q = DKGH$$

$$G = \frac{N}{K}$$

training $x_{FW} \rightarrow v_{model}$

inference

$$Q = X @ W_Q = BTKGH$$

$$K = X @ W_K = BSKH$$

$$V = X @ W_V = BSKH$$

$$X = BTD \rightarrow W_K = DKH$$

$$X = BSD \rightarrow W_V = DKH$$

$$W_O = DNH$$

$$Q @ K^T = BTKGH @ BSKH = B^2 TSKGH$$

$$\text{FLOPs} = 2B^2 TSKGH$$

$$\text{FLOPs} = B^2 TSKGH$$

$$S^2 (T=S)$$

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) @ V =$$

$$BTSKH @ BSKH = B^2 TSKGH$$

$$\text{FLOPs} = 2B^2 TSKGH$$

$$= B^2 TSKGH \quad (\text{Context } S)$$

Read

$$Q \rightarrow 2BTKGH$$

$$K \rightarrow 2BSKH$$

$$V \rightarrow 2BSKH$$

Write

$$O \rightarrow 2BTKGH$$

$$= 2BTN H$$

$$G = \frac{N}{K}$$

$$2BTKGH + 4BSKH + 2BTN H \rightarrow$$

$$AI = \frac{4BTskgH}{2BTkgH + 4BskgH + 2BTkH}$$

$$= \frac{TSg}{Tg + L}$$

$$T = L \quad (\text{training})$$

$$T \ll L$$

$$\frac{TSg}{Tg + L} \approx g$$

$$\frac{S^2g}{Sg + L} = \frac{Sg}{g + 1} \approx \frac{S}{2}$$

$$\begin{matrix} GQA \rightarrow \\ MQA \rightarrow \end{matrix} \quad GT \quad \boxed{G \approx N/k}$$

Training $\rightarrow AI \uparrow \rightarrow$ sequence length $\uparrow$

Inference $\rightarrow \boxed{AI \uparrow \rightarrow GT} \rightarrow (GQA, MQA)$

Question 5: At what sequence length are self-attention FLOPs equal to the QKVO projection FLOPs?

$$x \rightarrow [B, f, D]$$

$$G \approx N/k$$

$$Q \rightarrow D \times K \times H \rightarrow$$

$$\begin{bmatrix} B & T & D \\ x \end{bmatrix} \odot \begin{bmatrix} 0 & k & G & H \\ w_Q \end{bmatrix}$$

$$x \rightarrow [B, S, D] \rightarrow k \rightarrow \underline{D \times N \times H}$$

$$= 2 \text{ fwd} + 4 \text{ w.d.}$$

$$x \rightarrow [B, S, D] \rightarrow v \rightarrow \underline{D \times N \times H}$$

$$= 6 B T D K G H$$

$$(T=S)$$

$$Q \rightarrow 6 B T S K G H$$

$$K \rightarrow 6 B T S D N H$$

$$V \rightarrow 6 B T S D N H$$

$$O \rightarrow 6 B T S D N H$$

Attention 2

$$6 B T S K G H$$

+

$$6 B T S \frac{K G H}{N}$$

$$24 B T S D N H$$

$$12 B T S N H$$

$$= \boxed{20/k}$$

$$\boxed{20=S}$$



Question 6: Say we only save the output of each of the 7 main matmuls in a Transformer layer during our forward pass (Q, K, V, O + the three FFW matrices). How many extra FLOPs do we need to "rematerialize" during the backward pass?

Question 7: DeepSeek v3 says it was trained for 2.79M H800 hours on 14.8T tokens ([source](#)).

Given that it has 37B activated parameters, roughly what hardware utilization did they achieve?

Hint: note that they used FP8 FLOPs without structured sparsity.

$$2.79 \times 10^6 \text{ hours}$$

$$14.8 \times 10^{12} \text{ tokens}$$

$$37\text{B activated parameters}$$

$$1.513 \times 10^{15} \text{ FLOPs} \rightarrow \text{H800}$$

$$6 \times \text{num_params} \times \text{num_tokens}$$

$$\text{Total FLOPs} = 6 \times 37 \times 10^9 \times 14.8 \times 10^{12}$$

$$\text{time taken} = \frac{6 \times 37 \times 10^9 \times 14.8 \times 10^{12}}{1.513 \times 10^{15}}$$

$$\text{hardware utilization} = \frac{\frac{6 \times 37 \times 10^9 \times 14.8 \times 10^{12}}{1.513 \times 10^{15}}}{2.79 \times 10^6}$$

$$\frac{6 \times 37 \times 10^9 \times 14.8 \times 10^{12}}{1.513 \times 10^{15} \times 2.79 \times 10^6}$$

Question 8: Mixture of Experts (MoE) models have E copies of a standard dense MLP block, and each token activates k of these experts. What batch size in tokens is required to be compute-bound for an MoE with weights in int8 on TPU v5e? For DeepSeek, which has 256 (routed) experts and $k = 8$, what is this number?

Handwritten notes and diagrams illustrating the calculation for the batch size required for an MoE model to be compute-bound.

Diagram 1: A trapezoid representing the MoE structure. The top horizontal edge is labeled F (Experts). The bottom horizontal edge is labeled D (Tokens). A vertical line on the left is labeled k (Experts per token). A horizontal line across the middle is labeled k (Tokens per expert). A diagonal line is labeled k (Experts per token).

Diagram 2: A sequence of four boxes representing experts. The first box is labeled F and the third box is labeled D . Arrows point from the first and third boxes to a central circle containing a plus sign $(+)$. Below the circle is a box labeled D .

Calculation:

Input word
memory word
divide line

$$\frac{1.97e^{14}}{3.2e^{11}} \approx 210$$

f words $\rightarrow 2k \text{ DP } B$
Memory $\rightarrow GDF$

$$\frac{2kDFB}{GDF} = \frac{2kB}{G} > 210$$

$$B > \frac{120G}{k}$$

